Calcurion Physics Worksheets

A-Level Physics

Topic: Waves

MARKSCHEME

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1. Progressive: energy transferred; stationary: energy not transferred, nodes/antinodes **A2** [2]
2. Difference in phase between two points; radians (or degrees) **A2** [2]
3. Resultant displacement = sum of individual displacements **A2** [2]
4. $2A$ **A1** [1]
5. Constant phase difference **A1** [1]
6. Coherent sources; similar amplitude; path difference appropriate **A2** [2]
7. λ wavelength, D slit-to-screen, s slit separation **A3** [3]
8. $\lambda = 4.0 \times 10^{-7}$ m **M1A1** [3]
9. Spreading of wave when it passes through a gap or past an obstacle **A1** [1]
10. n order, a slit width **A1** [1]
11. $\theta \approx 0.17^\circ$ (or 3.0×10^{-3} rad) **M1A1** [3]
12. $n_1 \sin\theta_1 = n_2 \sin\theta_2$ **A1** [1]
13. $\theta_2 = 25.4^\circ$ **M1A1** [3]
14. Angle of incidence for which refraction angle $= 90^\circ$; $i > c$ **A2** [2]
15. $c = 41.8^\circ$ **M1A1** [2]
16. Incident and reflected waves superpose; fixed ends = nodes **A3** [3]
17. $\lambda = 1.6$ m; $v = 80$ m/s **M1A1** [3]
18. Path difference $= n\lambda$ for constructive **A3** [3]
19. $d = 2.0 \times 10^{-6}$ m; $\theta = 16.0^\circ$ **M1A1** [4]
20. Smaller $\lambda \rightarrow$ smaller w **A2** [2]
21. After first: $I_0/2$; after second: $(I_0/2)\cos^2 60^\circ = I_0/8$ **M1A1** [4]
22. $I = I_0 \cos^2\theta$ **A1** [1]
23. $I = I_0 \cos^2 30^\circ = 0.75 I_0$ **M1A1** [3]
24. Nodes: destructive interference (always zero); antinodes: constructive **A3** [3]
25. Measure θ for known order; $d\sin\theta = n\lambda$ **A4** [4]